

Rafif Ayyassar Wicaksono

Backend & AI/ML Engineer

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PROFILE

I am a backend and AI/ML engineer with 3+ years of professional experience (since 2022), working across fintech, telco, and enterprise. Over the last two years my focus has shifted to building and deploying Generative AI and backend systems — turning real business problems into working products rather than prototypes.

EXPERIENCE

Software Engineer — AI/ML & GenAI

Jun 2022 – Present

Freelance / Independent · concurrent with full-time roles

- Have delivered 20+ end-to-end AI/ML and backend projects for fintech, telco, and enterprise clients, from design through production deployment.
- Built and operate OprexDuit, my own personal-finance analytics SaaS (live in production, free, ~2,000 users) — Next.js/React frontend, secure backend with row-level security, and AI-powered spending analysis.
- Build LLM applications and automated agent workflows using RAG, Prompt Engineering, and n8n (with Vertex AI), for use cases such as customer chatbots, claims fraud detection, and natural-language data querying (NL2SQL).
- Build Python/FastAPI backends for AI inference, data processing, and multi-source REST integration.
- Deploy across multiple clouds — Google Cloud (primary, end-to-end), AWS, and Tencent Cloud / Alibaba Cloud for region-specific workloads — using Docker, Kubernetes, CI/CD, structured logging, and monitoring.
- Automate cross-system workflows with n8n, and use AI coding agents (Claude Code, Antigravity, GitHub Copilot) as part of the day-to-day development workflow.

Data Engineer — ETL

Aug 2025 – Apr 2026

Alfagift · retail e-commerce

- Automated data retention and cleanup across MongoDB, MySQL, and PostgreSQL via scheduled Python jobs.
- Implemented automated database and service upgrades on Cloud SQL, Redis, and Solr using Terraform (IaC).
- Built a cold-storage migration pipeline to GCS and BigQuery, reducing cloud storage cost by 20%.
- Optimized query performance using EXPLAIN ANALYZE and indexing strategies to speed up slow-running queries.
- Developed monitoring dashboards and alerting for database and pipeline health (Grafana, Cloud Operations).

Machine Learning Engineer

Dec 2024 – Mar 2025

Insignia (PT. Kreasi Media Asia) × Telkomsel · telco

- Developed and deployed Generative AI pipelines on Vertex AI for PDF summarization and metadata extraction.
- Built an AI web-scraping pipeline with OpenSearch and Gemini for structured content generation.

- Designed asynchronous AI workflows using AWS Lambda, S3, and GCP Cloud Functions; exposed ML services via REST APIs for enterprise integration.
- Applied MLOps with Vertex AI Pipelines, Docker, Kubernetes, and GitLab CI/CD.

Data Engineer — Operations

Aug 2024 – Aug 2025

Insignia (PT. Kreasi Media Asia) × Telkomsel · telco

- Monitored and supported batch and streaming pipelines using Apache Airflow and AWS CloudWatch (L1/L2).
- Optimized SQL and partitioning on PostgreSQL and Oracle to reduce data-warehouse cost.
- Managed ETL deployments via GitLab CI/CD with version control and rollback support.
- Implemented data-quality checks and anomaly detection, reducing manual QA effort by 40%.
- Collaborated with DS/ML teams to automate pipelines and maintain data lineage.

Earlier experience: Product Development (Contract) — FIF Group (Apr 2024 – Jul 2024); Product Development Intern — Adira Finance (Sep 2023 – Mar 2024); Data Analyst & Entry Intern — Bank Central Asia (BCA) (Dec 2022 – Sep 2023).

FEATURED PRODUCT

OprexDuit — Personal Finance Analytics SaaS

[🌐 oprexduit.vercel.app](https://oprexduit.vercel.app)

My own SaaS — a secure personal-finance analytics workspace to track transactions, budgets, and monthly insights, with AI-powered spending analysis. — ~2,000 users, Free to use, Live in production.

SELECTED PROJECTS

Insurance Chatbot — Registration & Claims Automation — Conversational assistant that lets customers register policies and file claims through chat.

Insurance / Fintech · LLM + RAG, n8n, Python/FastAPI, AWS Lambda & S3

Claim Fraud Detection — ML Pipeline — Machine-learning pipeline that scores insurance claims for fraud risk in real time.

Insurance / Fintech · Python (scikit-learn), FastAPI inference API, Model-performance monitoring

Finance Analyzer — Automated Financial Insights — AI tool that ingests financial data and produces summaries, charts, and insights automatically.

Fintech · Python/FastAPI, Data processing pipelines, LLM summarization, Automated visualization

Telco Customer Helpdesk Chatbot — AI helpdesk that handles customer complaints and service requests.

Telco · LLM intent classification, OpenSearch knowledge base, Internal ticketing integration, AWS

Ask Your Data — NL2SQL AI Engine — Natural-language interface that turns business questions into validated SQL with charts.

Enterprise · RAG + query validation, Python/FastAPI (secure execution), Auto-charting

Enterprise Complaint Management System — Fullstack web app for logging, routing, and analyzing complaints.

Enterprise · Python/FastAPI, Google Cloud Run, PostgreSQL, Firebase Auth

Internal Corporate Helpdesk Chatbot — Internal assistant for HR and operations requests and structured reporting.

Enterprise · n8n (orchestration), Vertex AI, Role-based access, OpenSearch indexing

HR & Employee Data Management API — Backend service for employee records and internal request management.

Enterprise · Python/FastAPI, Google Cloud Run, CI/CD, BigQuery

Smart Parking — Odd-Even License-Plate Detection — Undergraduate thesis: an automatic parking system that detects license plates and enforces odd-even rules using TensorFlow.

Academic / IoT & Computer Vision · TensorFlow, Computer Vision, Python, IoT

EDUCATION

Universitas Negeri Jakarta

B.Eng., Automation Engineering Technology
2020 – 2024

- GPA: 3.52 / 4.00.
- Thesis: automatic parking system for odd-even license-plate detection using TensorFlow.
- Academic projects: RFID smart door lock, smart parking system, heartbeat monitoring system, garbage-monitoring system.

TECHNICAL SKILLS

Generative AI & LLM: RAG, Prompt Engineering, Agent Workflows, n8n, MLOps, LLM evaluation, OpenAI GPT, Anthropic Claude, Google Gemini, Vertex AI, OpenSearch

LLM Frameworks & RAG: LangChain, LlamaIndex, LangGraph, Pydantic AI, MCP, pgvector, Qdrant, Langfuse, Ragas, vLLM, Ollama

AI-Assisted Development: Claude Code, Google Antigravity, GitHub Copilot, Cursor

Machine Learning: pandas, NumPy, Polars, scikit-learn, XGBoost, TensorFlow, Keras, PyTorch, MLflow, Vertex AI Pipelines

Backend & APIs: Python, FastAPI, REST APIs, Async processing, Microservices, Pydantic, SQLAlchemy, Shell scripting, C/C++

Data Engineering: Apache Airflow, dbt, Databricks, Spark / PySpark, Data Fusion, ETL/ELT, Batch & streaming, Data quality

Streaming & CDC: Apache Kafka, Redpanda, Debezium, Kafka Connect, Schema Registry, Apache Flink, GCP Pub/Sub

Databases: PostgreSQL (+ pgvector), MySQL, Oracle, SQL Server, MongoDB, Redis

Cloud Platforms: Google Cloud (primary), Cloud Run, BigQuery, Composer, Vertex AI, Cloud Functions, GCS, Firebase, Cloud SQL, AWS (Lambda, S3, CloudWatch, OpenSearch), Tencent Cloud, Alibaba Cloud

DevOps, CI/CD & IaC: Docker, Kubernetes, GitLab CI/CD, GitHub Actions, Azure DevOps, Terraform

Observability: Grafana, Datadog, AWS CloudWatch, Cloud Operations, Structured logging

BI & Visualization: Tableau, Power BI, Looker Studio, Excel